Charles Shaju

Embedded Systems Engineer

charlesshaju00@gmail.com | +91 9188210768 | charles-shaju.lovable.app

linkedin.com/in/charles-shaju | github.com/charles-shaju

SUMMARY

Embedded Systems Engineer with 2 years of experience in maritime robotics and IoT, currently pursuing an MCA to specialize in intelligent systems. Proven track record in developing Autonomous Surface Vessels (ASVs) and Underwater ROVs (UROV-SHM) using C/C++, ESP32, and Raspberry Pi. Expert in sensor fusion, real-time data systems, and UDP-based communication architectures for structural health monitoring

WORK EXPERIENCE

Embedded Systems Engineer — Aug/2023 – Jun/2025
i4 Marine Technologies Pvt Ltd — Pune, Maharastra
Full-time

- Experience in Embedded Systems Development (C/C++, ESP32, Raspberry pi)
- Worked on Underwater ROVs and Autonomous surface vessels (ASVs)
- Hands-on with Sensor Integration, Motor Control, LoRa, GPS, and real-time data systems

Robotics Intern — Jan/2023 – Jul/2023
Shristi Robotics — Thrissur

- Hands on building of Underwater ROV from Scratch
- Experience in 3D-Printing Technology
- Teaching various students on Mobile-Robotics

EDUCATION

MCA in Computer Application — Jun/2025 – Present
Rajagiri College of Social Science — Kalamassery, Kochi

Calicut University in Bachelore of Vocational Studies(IT) — Aug/2020 – Apr/2023
Christ College Autonomous Irinjalakuda — Thrissur
GPA: 85%

HSS(Plus Two) in Commerce Computer Application — Jul/2018 – Apr/2020
St. George HSS Pariyaram — Thrissur
GPA: 97%

SKILLS

Embedded Systems
ADC/DAC, Arduino, C / C++, Microcontroller Programming ESP32, PWM, STM32

Communication Protocols
I2C, LoRa, RS485/Modbus, SPI, UART, Wi-Fi

Sensors & Actuators
Barometric Sensors, Depth, GPS, IMU, Motor Control, pH, Pressure, Turbidity, Temperature

Systems & Platforms
BlueROV, Autonomous Surface Vessels, Raspberry Pi, MAVLink, MAVProxy, Pixhawk, QGroundControl, Underwater ROVS

Software Tools
Arduino IDE, Git, KiCad, Linux / Raspbian OS, Python, VS Code

ACHEIVEMENTS

Underwater Remotely Operated Vehicle (ROV) for Structural Health Monitoring
Indian Application No. 202241040473 A · 12 Jul 2024
UROV-SHM executes detailed underwater inspections of dams, tunnels, ship hulls, and bridge piers. Swappable batteries, fiber tethered telemetry, and AI-enhanced sonar/video analytics give real-time crack measurement, precise positioning, and mission resilience in turbulent conditions.

CERTIFICATIONS

Geo-Data Processing Using Python and ML
IIRS, ISRO

Certificate in AI-Machine Learning Developer
ASAP Kerala